Keele UNIVERSITY

Assessment Brief

Module Title	Advanced Applications of AI and ML	
Module Code	CSC-44112	
Assessment Type	Report	
Assessment Title	Assessment Part2: Technical Data Science Report	
Weighting (% of module mark)	70%	
Assessment Length (word count)	3500 words +/- 10% deviation	
Submission Deadline (date and time)	13/05/2026	1 pm
Format of Submission	Document as PDF and GitHub/Dataset Links	
Type of GenAI Use Permitted	Please refer to Use of AI section in this doc	
Feedback Release Date	03/06/2026	
Staff contact details	Dr Baidaa Al-Bander and Dr Sangeeta Sangeeta Appointment can be made through emails: b.al-bander@keele.ac.uk & s.sangeeta@keele.ac.uk	

Assessment Scope:

The second assessment component evaluates your ability to conduct a **complete end-to-end AI/data science project**, including: problem formulation, data preparation and analysis, model development and evaluation, interpretation of results, consideration of real-world impact, and professional communication of findings

Module Learning Outcomes:

In this assessment the following module learning outcomes will be assessed:

ILO1: Apply key concepts of Exploratory Data Analysis (EDA) to prepare and analyse datasets for machine learning.
ILO2: Assess the suitability of machine learning algorithms for various datasets and problem domains, considering computational efficiency and interpretability.
ILO3: Apply data science tools and libraries for data visualisation and machine learning to solve authentic problems.

Assessment Brief: Developing Technical Report (3500 words +/- 10% deviation**)**

You will complete a Technical Data Science Report based on a dataset of your own choosing. In this report, you will carry out a full data science investigation, including data preparation, analysis, modelling, and evaluation. You are expected to include appropriate visualisations and supporting figures, and to submit your code alongside the report, in order to demonstrate the practical skills, you have developed throughout the module. Your report should follow the structure and standards of a professional data science report, clearly communicating your methods, results, and conclusions.

Your Task: Following the report structure described below**,** develop a professional Data Science Report based on a dataset of your choice, demonstrating the application of advanced AI/ML techniques taught throughout the module.

Dataset Selection and Ethical Considerations: When selecting your dataset, choose one that is publicly available, ethically sourced, and appropriate for academic use. Avoid datasets that contain personal or sensitive information, such as identifiable personal data (names, addresses, IDs), medical records without proper anonymisation, or images that include recognisable human faces. You should also avoid social media data or scraped content unless it is clearly licensed for research use. Instead, select datasets that are clean, well-documented, and relevant to your research question. Make sure the dataset allows you to demonstrate meaningful data preparation, modelling, and evaluation without raising ethical, privacy, or legal concerns. If you are unsure about the ethical suitability of a dataset, please feel free to ask me for guidance before proceeding.

Report Structure

You must follow this structure exactly.

Report/Project Title

1. **Abstract (200 words)**

Concise summary of the problem, methods, results, and impact.

2. **Introduction and Problem Definition (400 words)**
   - Background, related work and motivation
   - Research question / problem statement
   - Aim and Objectives
3. **Exploratory Data Analysis (600 words)**
   - Dataset description
   - Data cleaning, preprocessing, feature visualisation/understanding and pattern discovery
   - Descriptive statistics, missing data analysis, etc
4. **Methodology (800 words)**
   - AI Model selection rationale
   - AI model implemented and architecture and pipeline design
   - Training and hyperparameter tuning
   - Tools and frameworks used
5. **Results and Evaluation (700 words)**
   - Testing and reporting evaluation metrics and evaluation strategy
   - Model comparison with other models in the literature
   - Output visualisations (e.g., confusion matrix, learning curves, visualisation of outputs, etc)
6. **Discussion and Real-World Impact (500 words)**
   - Interpretation of results and limitations
   - Practical implications, ethical, social and professional considerations
   - Deployment challenges
7. **Conclusion and Future Work (300 words)**
   - Key findings
   - Technical and practical lessons learned
   - Future improvements

8. References and Appendices (No limits)

Code link, dataset links, extended tables, hyperparameter logs, code snippets, additional plots, etc

Important Notes:

-You are expected to complete a set of programming tasks, including exploratory data analysis, data preprocessing, AI model development and training, as well as testing and evaluation. Each coding component will be assessed alongside the corresponding section in your report. For example, your Exploratory Data Analysis (EDA) code will be marked together with the EDA section of the report, which carries 20% of the total mark. The same approach applies to the other sections.

-Please make sure to clearly structure and label your code so that it matches the sections of your report. Use clear comments and headings in your notebook or script so I can easily identify which part of the code relates to each report section. Well-organised and clearly tagged code will make the marking process smoother and ensures that each component is assessed fairly alongside the relevant section of your report.

-Please note that simply providing the code is not sufficient. The outputs and results of your code must also be clearly shown and discussed in the report. For example, if you write code for Exploratory Data Analysis (EDA) but do not include the resulting figures, tables, or insights, this will negatively affect your mark. The same applies to model training and evaluation, if the training process, performance metrics, or visualisations (e.g., learning curves, confusion matrices, etc.) are not presented and interpreted, marks will be deducted. Your work must demonstrate both the implementation and the outcomes of your analysis.

Format of Submission

You will need to submit your work electronically by uploading it to blackboard using the link provided on KLE. Please save your document as PDF using the following format as the file name:

Student Number_Part2.pdf

Submit one PDF document containing your full report and any appendices, together with a **Public GitHub link** to your code and a **link or reference to the dataset** you used. The word limit for this assignment is 3500 words +/- 10% deviation **excluding** cover page, appendices, and references. Cover Page includes Module CSC-44112 Part 2, Student ID, Degree Programme, Academic Year, and Word Count. The references should be in Harward Reference style. You should follow this structure exactly:

- **Cover Page**
- **Project Title and Abstract (200 words)**
- **Introduction and Problem Definition (400 words)**
- **Exploratory Data Analysis (600 words)**
- **Methodology (800 words)**
- **Results and Evaluation (700 words)**
- **Discussion and Real-World Impact (500 words)**
- **Conclusion and Future Work (300 words)**
- **References and Appendices (No limits)**

Deadline & Time Management

Deadline:
The deadline for this is **1pm Wednesday 13th May 2026**

Late work will be penalised**.** All work submitted after the deadline in the absence of exceptional circumstances will be penalised, even if you manage to submit later that day. For work submitted as a first attempt, late work submitted within 7 days after the deadline will be capped at 50%. For work submitted as a second attempt, any submission after the deadline will be capped at 0%.

Marking Criteria

This assessment component informs 70% of your overall assessment. This assessment component will be marked out of 100 marks and then scaled to 70% for the overall portfolio. Marks will be awarded in line with the Keele generic marking guidelines (see the following summative rubric):

Section	Weight
Abstract	5%
Introduction and Problem Definition	10%
Exploratory Data Analysis	20%
Methodology	25%
Results and Evaluation	20%
Discussion and Real-World Impact	10%
Conclusion and Future Work	5%
References and Appendices	5%

Feedback to Students:

Feedback will be provided in the KLE module page within 3 weeks of submission, excluding periods of university closure.

Inclusive Practice:

This assessment has been explicitly designed to be inclusive in several ways:

1. Flexibility. Regular check-ins and support will be provided throughout the semester through weekly sessions. Several additional resources will be provided through lecture slides and reading material.
2. Exceptional circumstances (ECs). You can apply for an automatic, 7-day extension through the EC system as detailed below. Students with specific learning difficulties identified through Disability Support and Inclusion are not subject to the usual limit of 3 extension requests per semester. Alternatively, you can apply for another assessment opportunity, but will need appropriate supporting evidence of an unforeseen, acute event that is impacting your ability to engage with your studies. Detailed information on exceptional circumstances criteria, the claims process and evidence requirements can be found on the University web pages at: http://www.keele.ac.uk/ec/
3. Assistive technologies. We do allow the use of assistive technologies such as Grammarly/Generative AI when completing this assignment. Grammarly is a cloud-based programme that can be integrated into Microsoft Word and will make suggestions to rephrase sentences that will improve your writing. This can be particularly useful for students with dyslexia or students whose first language is not English.

Use of Artificial Intelligence (AI):

You are expected to learn and complete this assignment independently. GenAI can be used in this assessment as follows:

Type of AI Use	Description	Referencing Required
GenAI-assisted editing and proofreading	You may use GenAI to edit your work (e.g. enhancing phrasing, tone and sentence structure). You are not permitted to use GenAI to generate new content.	At the end of your work and before your reference list you must acknowledge your use of GenAI. For example: "*I acknowledge the use of Microsoft Copilot (version GPT-4, Microsoft, https://copilot.microsoft.com/) to edit my sentence structure and phrasing".*
Gen-AI-Assisted Reviewer	You may use GenAI to act as a reviewer to give academic feedback on a draft of your assessment (e.g. identifying areas for improvement). You are not permitted to use GenAI to generate new content.	At the end of your work and before your reference list you must acknowledge your use of GenAI. For example: "*I acknowledge the use of Microsoft Copilot (version GPT-4, Microsoft, https://copilot.microsoft.com/) to review a draft of my work and provide feedback".*
GenAI-Assisted Idea Generation	You may use GenAI to assist you to develop ideas (e.g., headings, bullet points, outline plans, basic structure), identify potential themes or relevant journal articles to assist you in the preparation of your assessment. However, you are not permitted to include GenAI generated content in your final assessment submission.	At the end of your work and before your reference list you must acknowledge your use of GenAI. For example, "*I acknowledge the use of Microsoft Copilot (version GPT-4, Microsoft, https://copilot.microsoft.com/) to identify key themes relevant to my presentation".*

Failure to disclose such use, if identified by the marker, will result in the matter being referred to the Academic Misconduct Officer (ACO) for appropriate action. You may then be required to attend an interview to demonstrate to the ACO that you have not misused these tools in completing the assessment. Please see Keele's Code of Practice on Academic Misconduct for further information.

Academic Misconduct:

Academic misconduct is doing something that could give you an unfair advantage in an assessment. It includes, but is not limited to, the following: plagiarism; collusion; contract cheating; cheating in an examination; falsification of data or sources; falsification of official documents or signatures. The University treats academic misconduct very seriously and penalties will be given for proven cases, including termination of studies in serious cases. It is therefore very important that you understand how to prepare and take assessments honestly. In order to assist you with this there are various resources and help available both as part of your programme of study and also centrally. For more information please visit: https://www.keele.ac.uk/students/academiclife/appeals-complaints-conduct/studentacademicconduct/

Academic Skills Support:

The Academic and Digital Skills team provide a range of additional online resources (e.g., study guides, Sways, Podcasts, workshops etc) to help you with your academic work and assessments. You can find more information here.